SUPERSEDING
Int. Fed. Spec. QQ-B-00825 (GSA-FSS)
July 1, 1963
Fed. Spec. QQ-C-504
March 31, 1947

FEDERAL SPECIFICATION

BUS BAR; COPPER, ALUMINUM, OR ALUMINUM-ALLOY

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers high-conductivity copper bar, rod, shapes, pipe, and tubing; aluminum-alloy bar, rod, shapes, and pipe; and electrical conductor grade (EC) aluminum bar for use as electrical conductors.

1.2 Classification.

1.2.1 *Types and classes.* Bus bars furnished under this specification shall be of the following types and classes, as specified (see 6.1).

Type I—High conductivity copper

Class 1—Bar
Class 2—Rod
Class 3—Shapes
Class 4—Pipe
Class 5—Tubing

Type II—Aluminum-alloy
Class 1—Bar
Class 2—Rod
Class 3—Shapes
Class 4—Pipe

Type III—Aluminum

Class 1—Bar

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government obtaining copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

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Military Specifications:

MIL-C-3993—Copper and Copper-Base Alloy Mill Products, Packaging of.

MIL-A-20695 — Aluminum Products, Preparation for Storage and Shipment of.

Military Standard:

MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following publications form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

***American Society for Testing and Materials Standards*:**

B187—Copper Bus Bar, Rod, and Shapes; Standard Specification for.

B188—Seamless Copper Bus Pipe and Tube; Standard Specification for.

B236—Aluminum Bars for Electrical Purposes (Bus Bars); Standard Specification for.

B317—Extruded Aluminum-Alloy Bar, Rod, Pipe, and Structural Shapes for Electrical Purposes; Standard Specification for.

(Copies may be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania. 19103.)

3. REQUIREMENTS

3.1 General.

3.1.1 *Type I bus bars.* Except as otherwise provided herein, type I bus bars shall conform to the requirements of ASTM B187 and B188, as applicable.

3.1.2 *Type II bus bars.* Except as otherwise provided herein, type II bus bars shall conform to the requirements of ASTM B317.

3.1.3 *Type III bus bars.* Except as otherwise provided herein, type III bus bars shall conform to the requirements of ASTM B236.

3.2 Materials. Unless otherwise specified (see 6.1), the copper for type I bus bars shall be of the ETP, OF, OFS, or STP classification.

3.3 Mechanical properties. Unless otherwise specified (see 6.1), tension, elongation, and bend tests shall be conducted on all sizes and forms of type I bus bars. Embrittlement tests shall be conducted on OF and OFS coppers only.

3.4 Edge contours. Unless otherwise specified (see 6.1), class 1 bus bars shall be finished with commercially square corners, as defined in the applicable ASTM standard.

4. SAMPLING, INSPECTION AND TEST PROCEDURES

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling, inspection, and tests of bus bars shall be conducted in accordance with the applicable ASTM standard to determine compliance with 4.1.

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4.3 **Rejection criteria.** If any of the tests fail to meet the requirements of this specification, the lot represented by the failed test specimen shall be rejected.

5. PREPARATION FOR DELIVERY

(For civil agency procurement, the definitions and applications of the various levels of packaging and packing shall be as specified in Fed. Std. No. 102.)

5.1 Preservation, packaging, and packing.

5.1.1 ***Type I bus bars.*** **The preservation, packaging, and packing of type I bus bars shall be in accordance with MIL-C-3993. Unless otherwise specified (see 6.1), level C shall be used.**

5.1.2 ***Types II and III bus bars.*** **The preservation, packaging, and packing of types II and III bus bars shall be in accordance with MIL-A-20695. Unless otherwise specified (see 6.1), level C shall be used.**

[redacted]

5.2.1 ***Civil agencies.*** **In addition to any special marking required by the contract or purchase order, shipping containers shall be marked in accordance with Fed. Std. No. 123.**

5.2.2 ***Military agencies.*** **In addition to any special marking required by the contract or purchase order, shipping containers shall be marked in accordance with MIL-STD-129.**

6. NOTES

6.1 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, symbol, and date of this specification.
(*b*) Type and class of bus bars (1.2.1).
(*c*) Kind of copper, if other than specified (3.2).
(*d*) Dimensions and form of bus bars.
(*e*) Temper of bus bar material.
(*f*) Elimination of type I physical tests, if desired (3.3).
(*g*) Class 1 bus bar edge contours, if other than specified (3.4).
(*h*) Quantity, in pounds or pieces.
(*i*) Inspection responsibility, if other than specified (4.1).
(*j*) Level of packing required (5.1.1 and 5.1.2).
(*k*) Special marking, if required (5.2).

6.2 Transportation description. Transportation descriptions and minimum weights applicable to this commodity are:

Rail:

Bus bars, aluminum.

Carload minimum weight 30,000 pounds.

[redacted]

Bus bars, aluminum.

Truckload minimum weight 30,000 pounds, subject to Rule 115, National Motor Freight Classification.

Rail:

Bus bars, copper.

Carload minimum weight 30,000 pounds.

Motor:

Bus bars, copper.

Truckload minimum weight 30,000 pounds, subject to Rule 115, National Motor Freight Classification.

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MILITARY INTEREST:

Military custodians:

Army—MO

Navy—WEPS

Air Force—ROAMA (88)

Review interest:

Army—MO

Navy—WEPS, DOCKS

Air Force—ROAMA (88)

User interest:

Navy—MC

Copies of this specification may be purchased for 5 cents each.

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